

SHWETA SINGH

Backend Engineer | Python | Distributed Systems

+91 8882291878 • shw12021982@gmail.com • Pune, India • [linkedin.com/in/shweta-singh-42b5a4380](https://www.linkedin.com/in/shweta-singh-42b5a4380) • <https://github.com/7007singh>

PROFESSIONAL SUMMARY

Backend Engineer with 3 years of experience building scalable backend systems, real-time data pipelines, and event driven microservices using Python, FastAPI, Kafka, Redis, and AWS. Proven ability to design distributed architecture processing 100K+ daily events, optimize API performance, and drive system design decisions for customer-facing platforms. Reduced database load by 60% through strategic caching and performance optimization.

TECHNICAL SKILLS

Backend & Frameworks Python, FastAPI, Django, Flask

Distributed Systems Kafka, ksqldb, Redis, Celery, Protocol Buffers, Message Queues

Databases & Search PostgreSQL, ClickHouse, MSSQL, OpenSearch, SQLAlchemy

Cloud & DevOps AWS (EC2, S3, Lambda, SQS), Docker, GitHub Actions

Engineering Concepts Microservices, Distributed Architecture, System Design, Event-Driven Architecture, REST APIs, Scalable Backend Systems, Caching, Async Processing, Performance Optimization, API Development, TDD

WORK EXPERIENCE

Software Engineer — Locobuzz Solutions Pvt. Ltd., Mumbai Aug 2023 – Jan 2026

Customer experience platform leveraging social media intelligence and analytics at scale.

- Designed and scaled FastAPI microservices for high-volume social media ingestion with Redis-based deduplication and Kafka event streaming, supporting high-availability burst loads across distributed pipelines.
- Engineered real-time data pipelines ingesting large-scale webhook traffic from Google My Business, Instagram, and other platforms using Kafka, Redis, and ClickHouse for low-latency analytics storage.
- Built a microservices alert system with 50+ dynamic rule configurations using Kafka + ksqldb for stream processing and Redis Sorted Sets for precision scheduling — processing 100K+ messages daily with near-zero message loss.
- Reduced database load by 60% through multi-layer caching with Redis and ClickHouse, delivering measurable improvements in API response times for customer-facing services.
- Optimized media retrieval performance via distributed caching and metadata indexing using AWS S3 and OpenSearch, reducing latency for high-frequency image assets.
- Packaged reusable Python libraries with CI/CD pipelines using GitHub Actions and Docker, accelerating deployment cycles and team productivity.

Software Developer Trainee — BridgeLabz Solutions LLP, Mumbai Dec 2022 – Jul 2023

- Developed full stack microservices projects (Fundoo Notes, Book Store) using Django, FastAPI, PostgreSQL, and Docker with end-to-end REST API integration.
- Achieved 90%+ test coverage using TDD with PyTest; documented all APIs in Postman and implemented Celery based async background workers.

KEY ACHIEVEMENTS

- High-Scale Alert Processing:** Designed an automated alert system processing 100,000+ messages per day with real-time and scheduled delivery, achieving near-zero message loss.
- Distributed Pipeline Architecture:** Delivered production-grade event-driven pipelines for 3+ social media platforms, enabling real-time scalable backend data ingestion at high throughput.

EDUCATION

B.Tech in Engineering — Delhi Institute of Tool Engineering, Delhi, India GPA: 8.1 / 10 | Jun 2022